

Visualization Index

Complete catalog of every generated figure in the project, organised by analysis stage. For detailed data-flow traces and downstream usage see [figure_trace_guide.md](#) (figure_trace_guide.md).

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Capacity Comparison	21	results/capacity_comparison/	capacity_sensitivity_analysis.py
Distance Comparison	4	results/distance_comparison/	run_distance_comparison_experiment.py
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Allocation Maps	3	results/maps/	Production pipeline
Statistical	1	results/figures/	Production pipeline
Heatmaps (parametric grid)	~160	results/heatmaps/	generate_capacity_sensitivity_heatmap.py
Total	~280		

1 Demand & EDA

Figure	Path	Description	Script / Notebook
fig_hourly_demand.png	results/figures/	24-hour crash demand profile	notebooks/02_eda_spatiotemporal.ipynb
fig_daily_demand.png	results/figures/	Day-of-week crash volume	notebooks/02_eda_spatiotemporal.ipynb
fig_temporal_trends.png	results/figures/	Long-term crash trends 2012-2026	notebooks/02_eda_spatiotemporal.ipynb
fig_crash_heatmap.png	results/figures/	Spatial heatmap of crash incidents	notebooks/02_eda_spatiotemporal.ipynb
fig_precinct_density.png	results/figures/	Precinct demand density choropleth	notebooks/02_eda_spatiotemporal.ipynb
fig_firehouses_map.png	results/figures/	Map of 48 Manhattan firehouses	notebooks/02_eda_spatiotemporal.ipynb
fig_cbd_comparison.png	results/figures/	CBD robustness comparison	notebooks/02_eda_spatiotemporal.ipynb
fig_precinct_demand.png	results/figures/	Precinct demand bar chart (original)	scripts/demand_modeling.py
pre-cinct_demand_rates.png	results/figures/	Basic precinct demand rates bar chart	scripts/generate_precinct_demand_visualizations.py
pre-cinct_demand_rates_improved.png	results/figures/	Improved bar chart with legend, annotations	scripts/generate_precinct_demand_visualizations.py
pre-cinct_demand_heatmap.png	results/figures/	Spatial heatmap of precinct demand	scripts/generate_precinct_demand_visualizations.py
fig_demand_model_fit.png	results/figures/	NHPP model fit diagnostics	scripts/demand_modeling.py
	results/figures/		

Figure	Path	Description	Script / Notebook
fig_hourly_rates.png		CBD vs non-CBD lambda factors	scripts/demand_modeling.py
temporal_demand_patterns.png	results/figures/	Temporal demand pattern overview	Production pipeline

2 Service & Travel Models

Figure	Path	Description	Script / Notebook
distance_matrix_heatmap.png	results/figures/	Haversine distance matrix heatmap	notebooks/ 04_service_travel_proxy.ipynb
distance_matrix_coverage.png	results/figures/	Distance matrix coverage visualization	notebooks/ 04_service_travel_proxy.ipynb
travel_time_by_tod.png	results/figures/	Travel time by time- of-day band	notebooks/ 04_service_travel_proxy.ipynb
tod_speed_factors.png	results/figures/	24-hour speed factor profile	notebooks/ 04_service_travel_proxy.ipynb
service_time_distribution.png	results/figures/	LogNormal service time distribution	notebooks/ 04_service_travel_proxy.ipynb
nhpp_arrivals_demo.png	results/figures/	NHPP arrival genera- tion demo	notebooks/ 04_service_travel_proxy.ipynb

3 Optimization

Figure	Path	Description	Script / Notebook
fig_policy_comparison.png	results/figures/	Policy comparison (P0, P1, P2)	scripts/run_optimization_comparison.py
fig_tradeoff_curve.png	results/figures/	RT vs coverage trade-off	scripts/run_optimization_comparison.py
opt_allocation_comparison.png	results/figures/	Allocation maps by model	scripts/run_optimization_comparison.py
opt_inputs.png	results/figures/	Optimization inputs visualization	scripts/run_optimization_comparison.py
opt_sensitivity.png	results/figures/	Objective vs fleet size	scripts/run_optimization_comparison.py
policy_comparison.png	results/figures/	Alternative policy comparison chart	Production pipeline
allocation_heatmaps.png	results/figures/optimization/	Optimization allocation heatmaps	scripts/run_optimization_comparison.py
policy_comparison.png	results/figures/optimization/	Optimization-specific policy comparison	scripts/run_optimization_comparison.py

4 Simulation Experiments

Figure	Path	Description	Script / Notebook
exp1_policy_comparison.png	results/figures/	Experiment 1: Policy box plots	notebooks/07_production_results.ipynb
exp2_fleet_sensitivity.png	results/figures/	Experiment 2: RT vs fleet size	notebooks/07_production_results.ipynb
exp3_demand_sensitivity.png	results/figures/	Experiment 3: RT vs demand multiplier	notebooks/07_production_results.ipynb
exp4_service_robustness.png	results/figures/	Experiment 4: RT vs service time	notebooks/07_production_results.ipynb

5 Verification & Validation

Figure	Path	Description	Script / Notebook
verification_toy_timeline.png	results/figures/	Toy example timeline	notebooks/06_simulation_debug.py
validation_p0_vs_p2.png	results/figures/	Pilot 1: P0 vs P2	notebooks/06_simulation_debug.py
validation_sensitivity_K.png	results/figures/	Pilot 2: RT vs K	notebooks/06_simulation_debug.py
validation_sensitivity_demand.png	results/figures/	Pilot 3: RT vs demand	notebooks/06_simulation_debug.py

6 Extended Fleet Analysis

6a Summary figures (results/figures/)

Figure	Path	Description	Script / Notebook
fleet_sensitivity_dual.png	results/figures/	Dual-panel fleet sensitivity	scripts/ run_production_v2.py
fleet_sensitivity_curve.png	results/figures/	Fleet sensitivity single curve	scripts/ run_production_v2.py
fleet_sensitivity_v2_dual.png	results/figures/	V2 dual-panel fleet sensitivity	scripts/ run_production_v2.py
production_fleet_sensitivity.png	results/figures/	Production fleet sensitivity chart	scripts/ run_production_v2.py
demand_sensitivity_curve.png	results/figures/	Demand sensitivity curve	Production pipeline
response_time_distribution_by_policy.png	results/figures/	RT distribution by policy	Production pipeline
policy_comparison_panel_K20_cap2.png	results/figures/	Staging map panel (P0/P1/P2)	Production pipeline
allocation_comparison_K20.png	results/figures/	Allocation comparison for K=20	Production pipeline

6b Extended fleet detail ([results/production_v2/figures/](#))

Figure	Path	Description	Script / Notebook
allocation_map_K20.png	results/production_v2/figures/	Allocation map at K=20	scripts/run_production_v2.py
allocation_map_K30.png	results/production_v2/figures/	Allocation map at K=30	scripts/run_production_v2.py
allocation_map_K40.png	results/production_v2/figures/	Allocation map at K=40	scripts/run_production_v2.py
coverage_vs_K.png	results/production_v2/figures/	Coverage vs fleet size	scripts/run_production_v2.py
effect_sizes.png	results/production_v2/figures/	Statistical effect sizes	scripts/run_production_v2.py
mean_rt_vs_K.png	results/production_v2/figures/	Mean response time vs fleet size	scripts/run_production_v2.py
p95_rt_vs_K.png	results/production_v2/figures/	P95 response time vs fleet size	scripts/run_production_v2.py
queue_metrics_vs_K.png	results/production_v2/figures/	Queue metrics vs fleet size	scripts/run_production_v2.py
rt_distribution_K20.png	results/production_v2/figures/	RT distribution at K=20	scripts/run_production_v2.py
rt_distribution_K30.png	results/production_v2/figures/	RT distribution at K=30	scripts/run_production_v2.py
rt_distribution_K40.png	results/production_v2/figures/	RT distribution at K=40	scripts/run_production_v2.py
utilization_vs_K.png	results/production_v2/figures/	Unit utilization vs fleet size	scripts/run_production_v2.py

7 CBD Analysis

7a Main CBD figures (results/figures/)

Figure	Path	Description	Script / Notebook
cbd_heatmap.png	results/figures/	CBD crash density + firehouses	notebooks/09_cbd_analysis.ipynb
cbd_response_comparison.png	results/figures/	CBD stress RT comparison	notebooks/09_cbd_analysis.ipynb
cbd_scenario_comparison.png	results/figures/	CBD vs Manhattan scenarios	notebooks/09_cbd_analysis.ipynb
cbd_robustness.png	results/figures/	CBD robustness analysis (basic)	Production pipeline
cbd_robustness_enhanced.png	results/figures/	Enhanced CBD robustness	Production pipeline
cbd_equity_tradeoff_summary.png	results/figures/	CBD equity-efficiency trade-off summary	scripts/run_cbd_focused_optimization.py

7b CBD focused comparison (results/cbd_focused_comparison/)

Figure	Path	Description	Script / Notebook
cbd_focused_comparison.png	results/cbd_focused_comparison/	CBD-focused RT comparison bar chart	scripts/run_cbd_focused_optimization.py
allocation_comparison.png	results/cbd_focused_comparison/	CBD allocation distribution comparison	scripts/run_cbd_focused_optimization.py
equity_tradeoff.png	results/cbd_focused_comparison/	CBD vs non-CBD equity trade-off	scripts/run_cbd_focused_optimization.py

8 Queue Analysis

Figure	Path	Description	Script / Notebook
queue_comparison_by_policy.png	results/figures/	Queue metrics by policy	scripts/analyze_queue_metrics.py
queue_heatmap.png	results/figures/	Queue length heatmap	scripts/analyze_queue_metrics.py
queue_metrics.png	results/figures/	Queue metrics overview	scripts/analyze_queue_metrics.py
queue_vs_demand.png	results/figures/	Queue vs demand multiplier	scripts/analyze_queue_metrics.py
queue_vs_fleet_size.png	results/figures/	Queue vs fleet size	scripts/analyze_queue_metrics.py

9 Seasonal Analysis

Figure	Path	Description	Script / Notebook
seasonal_patterns.png	results/figures/	Seasonal variation analysis	scripts/analyze_seasonal_patterns.py
seasonal_decomposition.png	results/figures/	Seasonal decomposition	scripts/analyze_seasonal_patterns.py
seasonal_heatmap.png	results/figures/	Month x DoW heatmap	scripts/analyze_seasonal_patterns.py

10 Capacity Sensitivity

10a Summary figures (results/figures/)

Figure	Path	Description	Script / Notebook
capa-city_sensitivity_heatmap.png	results/figures/	RT by policy and capacity	scripts/generate_capacity_sensitivity_heatmap.py
capa-city_sensitivity_heatmap_notebook.png	results/figures/	RT heatmap (notebook variant)	Notebook pipeline
fire-house_capacity_analysis.png	results/figures/	Capacity analysis detail	scripts/capacity_sensitivity_analysis.py

10b Capacity comparison detail ([results/capacity_comparison/](#))

Figure	Path	Description	Script / Notebook
allocation_comparison.png	results/capacity_comparison/	Allocation comparison overview	scripts/capacity_sensitivity_analysis.py
allocation_comparison_K20.png	results/capacity_comparison/	Allocation comparison at K=20	scripts/capacity_sensitivity_analysis.py
allocation_comparison_K40.png	results/capacity_comparison/	Allocation comparison at K=40	scripts/capacity_sensitivity_analysis.py
cbd_distribution.png	results/capacity_comparison/	CBD unit distribution	scripts/capacity_sensitivity_analysis.py
cbd_distribution_K20.png	results/capacity_comparison/	CBD distribution at K=20	scripts/capacity_sensitivity_analysis.py
cbd_distribution_K40.png	results/capacity_comparison/	CBD distribution at K=40	scripts/capacity_sensitivity_analysis.py
concentration_analysis.png	results/capacity_comparison/	Unit concentration analysis	scripts/capacity_sensitivity_analysis.py
concentration_analysis_K20.png	results/capacity_comparison/	Concentration at K=20	scripts/capacity_sensitivity_analysis.py
concentration_analysis_K40.png	results/capacity_comparison/	Concentration at K=40	scripts/capacity_sensitivity_analysis.py
full_spectrum_summary.png	results/capacity_comparison/	Combined 6-panel summary figure	scripts/capacity_sensitivity_analysis.py
max_units_vs_capacity_K20.png	results/capacity_comparison/	Max units per firehouse at K=20	scripts/capacity_sensitivity_analysis.py
max_units_vs_capacity_K40.png	results/capacity_comparison/	Max units per firehouse at K=40	scripts/capacity_sensitivity_analysis.py

Figure	Path	Description	Script / Notebook
performance_comparison.png	results/capacity_comparison/	Performance comparison overview	scripts/capacity_sensitivity_analysis.py
performance_comparison_K20.png	results/capacity_comparison/	Performance at K=20	scripts/capacity_sensitivity_analysis.py
performance_comparison_K40.png	results/capacity_comparison/	Performance at K=40	scripts/capacity_sensitivity_analysis.py
performance_vs_capacity_K20.png	results/capacity_comparison/	Performance metric curves at K=20	scripts/capacity_sensitivity_analysis.py
performance_vs_capacity_K40.png	results/capacity_comparison/	Performance metric curves at K=40	scripts/capacity_sensitivity_analysis.py
rt_heatmap_K20.png	results/capacity_comparison/	Policy x capacity RT heatmap at K=20	scripts/capacity_sensitivity_analysis.py
rt_heatmap_K40.png	results/capacity_comparison/	Policy x capacity RT heatmap at K=40	scripts/capacity_sensitivity_analysis.py
tradeoff_dispersion_rt_K20.png	results/capacity_comparison/	Firehouses used vs mean RT at K=20	scripts/capacity_sensitivity_analysis.py
tradeoff_dispersion_rt_K40.png	results/capacity_comparison/	Firehouses used vs mean RT at K=40	scripts/capacity_sensitivity_analysis.py

11 Distance Comparison

Figure	Path	Description	Script / Notebook
<code>dis-distance_matrices_heatmap.png</code>	<code>results/dis-distance_comparison/</code>	Side-by-side distance heatmaps	<code>scripts/run_distance_comparison_experiment.py</code>
<code>dis-distance_scatter.png</code>	<code>results/dis-distance_comparison/</code>	Haversine vs Manhattan scatter	<code>scripts/run_distance_comparison_experiment.py</code>
<code>dis-distance_comparison_bar.png</code>	<code>results/dis-distance_comparison/</code>	Performance bar chart	<code>scripts/run_distance_comparison_experiment.py</code>
<code>dis-distance_comparison_boxplot.png</code>	<code>results/dis-distance_comparison/</code>	Replication box plots	<code>scripts/run_distance_comparison_experiment.py</code>

12 P0 Spatial Baseline

Figure	Path	Description	Script / Notebook
<code>p0_spatial_map.png</code>	<code>results/figures/</code>	P0 (spatially-stratified) firehouse selections	<code>scripts/p0_spatial_analysis.py</code>
<code>p0_spatial_metrics.png</code>	<code>results/figures/</code>	Stratification comparison metrics	<code>scripts/p0_spatial_analysis.py</code>
<code>p0_spatial_north_south.png</code>	<code>results/figures/</code>	North-south distribution	<code>scripts/p0_spatial_analysis.py</code>
<code>p0_vs_p2_response_time.png</code>	<code>results/figures/</code>	P0 vs P2 response time comparison	<code>scripts/p0_spatial_analysis.py</code>

13 Publication-Quality Figures

Figure	Path	Description	Script / Notebook
pub_fig1_policy_comparison.png	results/figures/	Publication: Policy comparison	scripts/generate_publication_figures.py
pub_fig2_fleet_sensitivity.png	results/figures/	Publication: Fleet sensitivity	scripts/generate_publication_figures.py
pub_fig3_demand_robustness.png	results/figures/	Publication: Demand robustness	scripts/generate_publication_figures.py
pub_fig4_service_sensitivity.png	results/figures/	Publication: Service sensitivity	scripts/generate_publication_figures.py
pub_fig5_performance_heatmap.png	results/figures/	Publication: Performance heatmap	scripts/generate_publication_figures.py

14 Response Time Trade-off

Figure	Path	Description	Script / Notebook
re-response_time_coverage_tradeoff.png	results/figures/	RT vs coverage trade-off (original)	scripts/generate_publication_figures.py
re-response_time_coverage_tradeoff_improved.png	results/figures/	Improved RT vs coverage with data table	scripts/generate_publication_figures.py
re-response_time_coverage_tradeoff_zoomed.png	results/figures/	Zoomed high-performance region	scripts/generate_publication_figures.py

15 Statistical Analysis

Figure	Path	Description	Script / Notebook
statistical_effect_sizes.png	results/figures/	Effect sizes across experiments	Production pipeline

16 Allocation Maps

Figure	Path	Description	Script / Notebook
map_allocation_P0_K40.png	results/maps/	P0 allocation map at K=40	Production pipeline
map_allocation_P1_K40.png	results/maps/	P1 allocation map at K=40	Production pipeline
map_allocation_P2_K40.png	results/maps/	P2 allocation map at K=40	Production pipeline

17 Dashboard

Figure	Path	Description	Script / Notebook
project_summary_dashboard.png	results/figures/	Full project summary	scripts/generate_summary_dashboard.py

18 Heatmaps (Parametric Grid)

Systematic allocation heatmaps covering all combinations of fleet size, policy, and capacity.

Location: results/heatmaps/

Naming convention: heatmap_K{k}_policy{policy}_cap{capacity}.png

- **Fleet sizes (K):** 5, 10, 15, 20, 25, 30, 35, 40, 45
- **Policies:** P0, P0_spatial, P1, P2
- **Capacities:** 1, 2, 3, 5

Example	Description
<code>heatmap_K20_policyP2_cap2.png</code>	Optimized allocation (P2), 20 units, capacity 2
<code>heatmap_K10_policyP0_spatial_cap1.png</code>	Spatial baseline (P0), 10 units, capacity 1
<code>heatmap_K40_policyP1_cap5.png</code>	Demand-proportional (P1), 40 units, capacity 5

Corresponding allocation CSVs are stored in `results/heatmaps/allocations/`.

Regeneration

All figures can be regenerated from source data. See [figure_trace_guide.md](#) (`figure_trace_guide.md`) for the full list of regeneration commands.